Saveetha Institute of Medical and Technical Sciences
SAVEETHA SCHOOL OF ENGINEERING

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CO3-AT1- EXPERIMENTS

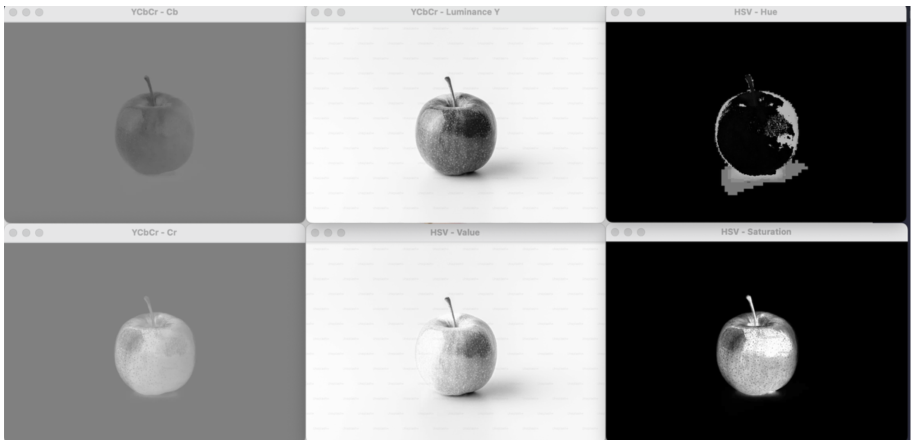
Module 1: Image Representation, Types of Image Processing & Benefits

Exercise 2: Color Space Robustness under Ambient Drift

- Real-Time Scenario: A camera is tracking an object under variable room lighting. When a shadow passes over the room, a standard RGB pixel representation changes values wildly, causing tracking failures.
- Task:
- Explain why the linear formula $Y = 0.299R + 0.587G + 0.114B$ is used to compress color data into a grayscale baseline rather than a simple arithmetic average.
- Propose a preprocessing workflow that splits the color channels into an illumination-independent space (such as HSV or YCbCr). Explain how isolating the luminance/value component (Y or V) from the chrominance components allows the vision pipeline to remain invariant to sudden ambient lighting changes.

Input Image

Output Images



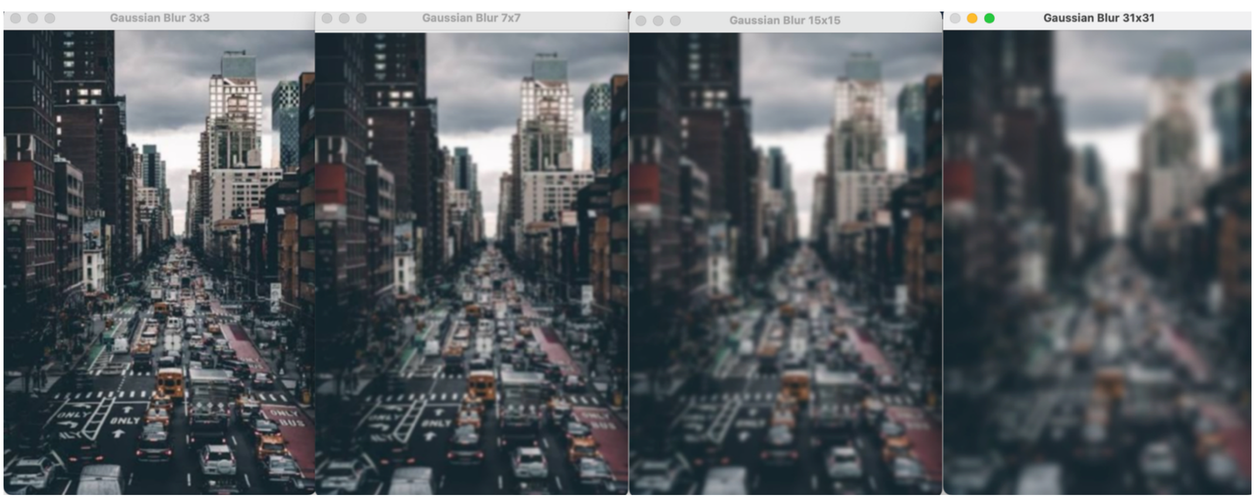
Module 2: Image Preprocessing (Linear vs. Non-Linear Filtering)

Exercise 4: Kernel Resolution vs. Computational Performance

- Real-Time Scenario: You are applying a real-time low-pass blurring filter to smooth an image before running a feature detector.
- Task:
- Track the frame rate (FPS) of the pipeline while scaling the filter kernel matrix size across four steps: 3 × 3, 7 × 7, 15 × 15, and 31 × 31.
- Quantify the exact relationship between spatial kernel scaling and processing lag.
- Using Big-O spatial convolution complexity notation, explain why a massive linear kernel degrades real-time responsiveness, and identify at which kernel threshold the frame rate drops below an acceptable threshold (<15 FPS).

Input Image

Output Images



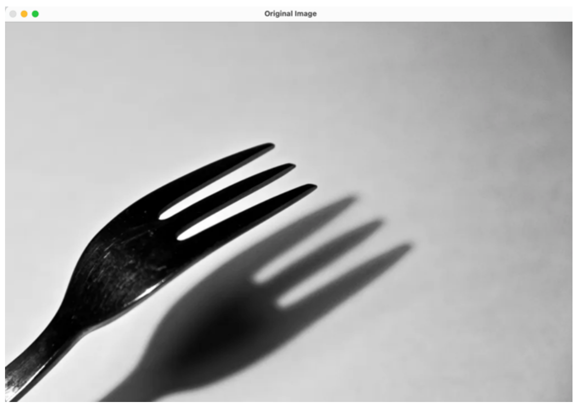
Module 3: Edge Detection (Spatial Gradient Analytics)

Exercise 6: Dynamic Hysteresis Tuning under Low Contrast

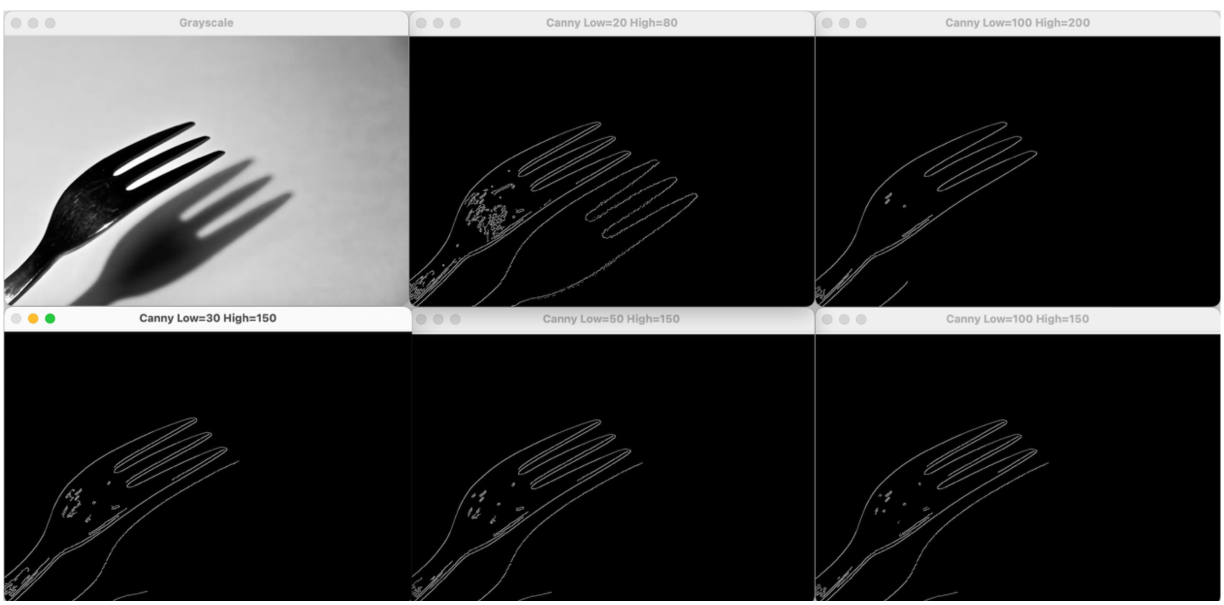
- Real-Time Scenario: You are attempting to isolate object silhouettes using a Canny edge pipeline. The object has high contrast against the background on one side, but a soft, fading shadow on the other.
- Task:

- Manually adjust the dual-threshold parameters (High Threshold and Low Threshold) of the hysteresis system.
- Document what happens when the gap between the high and low thresholds is too wide versus too narrow.
- Detail how the edge tracker uses "edge linking" to bridge faint boundaries in low-contrast zones, and identify the point where tuning results in either broken edges or a flood of false-positive noise loops. [1, 2, 3]

Input Image



Output Images



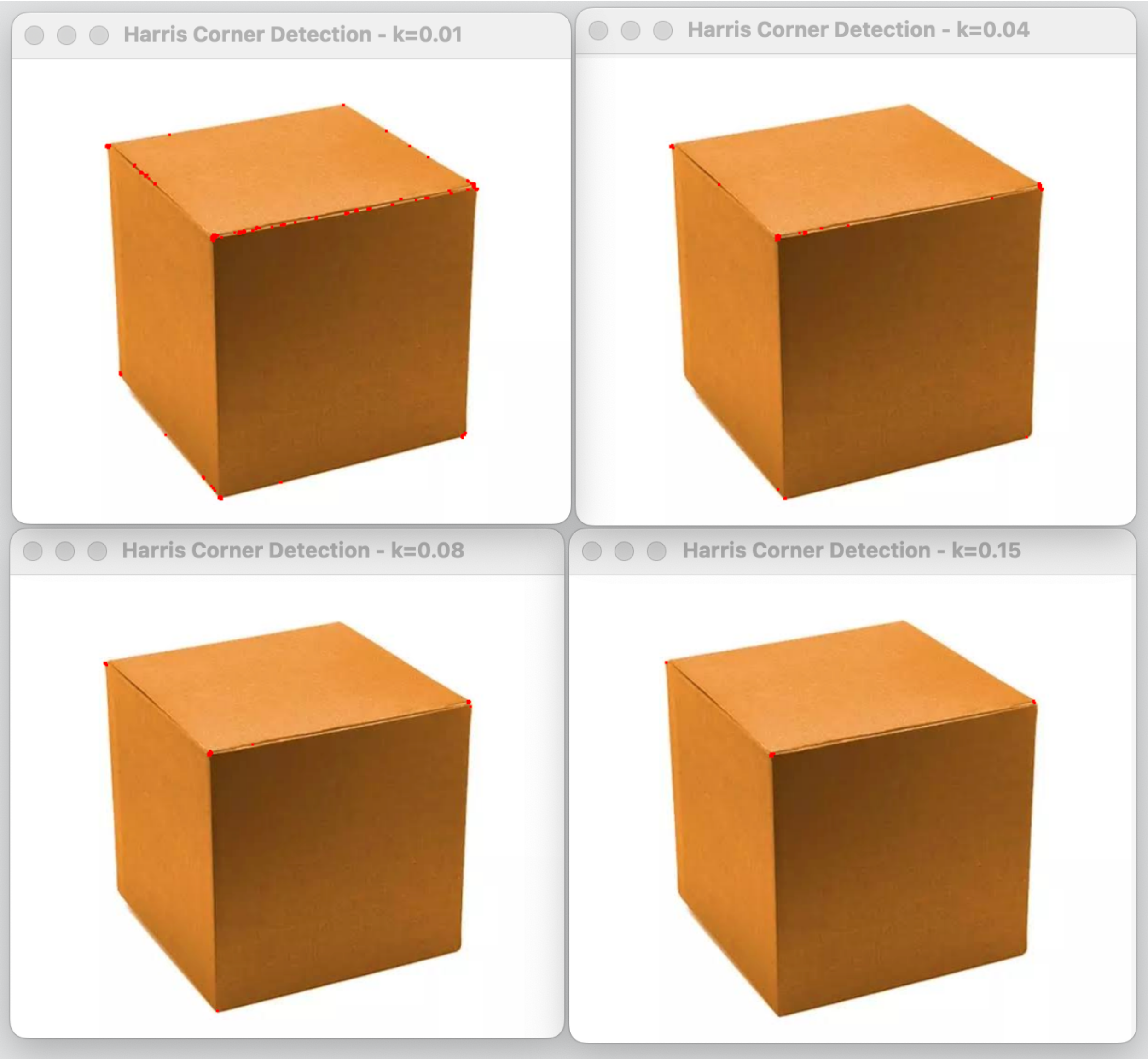
Module 4: Corner Detection (Harris Spatial Tensor Analytics)

Exercise 8: Sensitivity Engineering via the Free Parameter (k)

- Real-Time Scenario: You are tracking corner features on a metallic object that reflects light, creating false glints that look like features.
- Task:
- Locate the free empirical scaling constant k inside the Harris response formula: $R=\text{Det}(M)-k(\text{Trace}(M))^{2}$
- Systematically modify the value of k from a highly sensitive state (k = 0.01) to a conservative state (k = 0.15).
- Document how shifting k penalizes the trace of the matrix. Describe the trade-off this parameter controls between suppressing weak, noise-induced false corners and retaining valid, low-contrast structural corners.

Input Image

Output Images



Result

The experiments demonstrated the effects of **color-space conversion, kernel size, Canny hysteresis thresholds, and the Harris parameter** on image-processing performance. The results showed that appropriate parameter selection improves **image quality, edge continuity, corner detection accuracy, noise suppression, and real-time processing efficiency**.